Bangladesh Army University of Engineering & Technology

Department of Computer Science and Engineering

B.Sc. Eng. 1st Year 2nd Semester, Semester Final Examination, Fall-2023

Course Code: GED-1251 | **Course Title: Society, Ethics and Environment**

Full Marks: 70 | **Time: 03 Hours**

N.B.:

1. Figures shown in the right margin indicate full marks.
2. Answer **Six** questions, taking **Three** from each section **including Q.1, Q.2, Q.5 and Q.6**
3. Use separate answer script for each section.
4. Assume reasonable data if necessary.

Section-A

Q.1 (a) Ethics are crucial in engineering. They not only promote quality work but also encourage people to operate safely and maintain a high standard of ethical responsibility. Why do you need to study engineering ethics? Discuss. **[CO1, PO1, BL1, KP1]** 3

(b) The essence of engineering practice is the design of products. Describe ethics problems are like design problem with examples. **[CO2, PO6, BL2, KP1]** 3

(c) Engineers often encounter situations in which they don't have all the information that is needed. Explain engineering is managing the unknown with the disaster at Bhopal case study. **[CO3, PO6, BL2, KP1]** 6

Q.2 (a) The astronauts on the *Challenger* mission were aware of the dangerous nature of riding a complex machine such as the space shuttle, so they can be thought of as having given informed consent to participating in a dangerous enterprise. What role did inform consent play in this case? **[CO2, PO6, BL1, KP1]** 3

(b) Do you think that the astronauts had enough information to give informed consent to launch the shuttle that day? **[CO2, PO6, BL2, KP1]** 3

(c) Moral theory defines terms in uniform ways and links ideas and problems together in consistent ways. Utilitarianism is one of the moral theories that emphasizes the well-being of society as a whole. Illustrate the idea of utilitarianism theory shortly. **[CO3, PO7, BL4, KP1]** 6

Q.3 (a) Describe shortly the necessity of ethical principles of the IEEE? 3

(b) To maintain and improve our technical competence and to undertake technological tasks for others only if qualified by training or experience, or after full disclosure of pertinent limitations. Explain IEEE codes of ethics and IEEE code of conduct. 4

(c) Why is it important that engineers work ethically in relation to honesty and integrity? Discuss. 4

Q.4 (a) What is Environmental policy and protection tools? Clarify. 5

(b) "Environmental policy is stitching together the fabric of sustainability for generations to come." Explain key environmental policy and protection tools with examples in Bangladesh. 6

Section-B

Q.5 (a) One tool often used in engineering analysis, especially when trying to determine whether a project makes sense is cost-benefit analysis. Do you agree that cost-benefit analysis is just an application of

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utilitarianism? Discuss. **[CO2, PO6, BL2, KP1]** 4

(b) Identify what does utilitarianism tells us about the behaviour of the Aberdeen Three. **[CO3, PO8, BL2, KP1]** 4

(c) Distinguish between duty rights and ethics rights with examples. 4

Q.6 (a) Distinguish between engineering and other professions with example. **[CO1, PO1, BL2, KP1]** 3

(b) One objection to codes of ethics is the internal conflicts that can exist within them, with no instructions on how to resolve this conflict. Can codes resolve internal conflict within engineers? **[CO2, PO6, BL1, KP1]** 4

(c) Explain how codes and professional societies protect employees? Give an example from the Intel Pentium Chip case study. **[CO3, PO8, BL2, KP1]** 5

Q.7 (a) Why is sustainable development so often associated with protecting the environment? 3

(b) Identify the major issues of Sustainable Development Goals? Is it relevant to engineers? Discuss. 4

(c) Illustrate the relations between sustainable development goals and the prevention of environmental pollution in Dhaka city. 4

Q.8 (a) Define cyber security and classify the domains of cyber security. 3

(b) Illustrate at least five advantages or benefits of cyber security with examples. 4

(c) Describe the relationship between the internet and free speech in the era of social media. 4

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